

B.Sc. BOTANY — YEAR I

MAJOR COURSE III

APPLIED BOTANY

UNIT III — Agriculture, Horticulture, Forestry and Silviculture

Detailed Teacher-Style Notes

Definitions • Concepts • Tables • Flow diagrams • Examples • Applications • Exam points

Prepared in the same detailed format as the Major Course II notes.

UNIT III — AGRICULTURAL AND LAND-USE PRACTICES

Syllabus topics covered: ancient agricultural practices; modern agriculture—polyhouse, drip irrigation, hydroponics, computer-based agriculture and terrace farming; organic farming; horticulture; forestry; silviculture; ancient fertilizers and pesticides.

3.1 Ancient agricultural practices

Traditional agriculture commonly relied on local ecological knowledge, crop diversity, organic residues, livestock-linked nutrient cycling and seasonal management. A scientifically strong answer should separate documented practice from unverified claims and explain the underlying agronomic principle.

Practice principle	Agronomic value
Mixed cropping	Risk distribution and possible better resource use
Organic residues	Return of carbon and nutrients to soil
Seed selection	Maintenance of locally adapted planting material
Water conservation	Better use of limited water
Local seasonal knowledge	Alignment of cultivation with climate patterns

3.2 Modern agriculture practices

Polyhouse

A protected cultivation structure covered with suitable material that modifies the crop microclimate. It can support off-season production and protection, but requires capital, management and ventilation.

Drip irrigation

A micro-irrigation method that delivers water near the root zone through a network of pipes and emitters. It can improve water-use efficiency when designed and maintained correctly.

Hydroponics

Cultivation of plants without conventional soil as the main rooting medium, using a carefully managed nutrient solution and suitable support system.

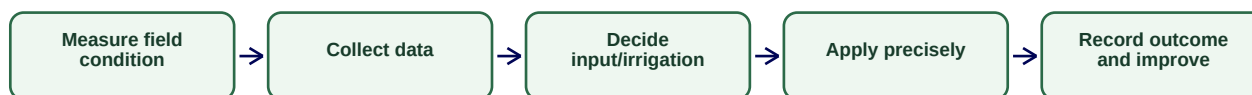
Computer-based agriculture

Use of sensors, records, mapping, decision-support tools, weather data and automation to improve farm decisions.

Terrace farming

Cultivation on step-like land surfaces, especially in hilly areas, to reduce slope erosion and make land more manageable.

Logic of computer-assisted / precision agriculture



3.3 Organic farming

Definition — Organic farming: A production system that emphasizes ecological processes, biodiversity, recycling of organic resources and restricted reliance on synthetic external inputs according to applicable standards.

Component	Role
Compost and organic matter	Supports soil structure and nutrient cycling
Crop rotation	Breaks repeated crop cycles and diversifies production
Biological pest management	Uses ecological and biological approaches

Component	Role
Soil biodiversity	Supports decomposition and nutrient transformations
Certification/standards	Defines permitted practices for marketed organic products

3.4 Horticulture

Definition — Horticulture: The branch of agriculture concerned with the cultivation, improvement, production and management of fruits, vegetables, flowers, ornamentals and related crops.

- **Objectives:** quality produce, intensive cultivation, post-harvest value and income generation.
- **Techniques:** nursery management, propagation, pruning, protected cultivation, irrigation and post-harvest handling.

3.5 Forestry

Definition — Forestry: The science, art and practice of establishing, managing, conserving and using forests and tree resources for ecological, economic and social objectives.

Forestry supports timber and non-timber products, wildlife habitat, watershed protection, carbon storage and livelihoods. Sustainable forestry requires regeneration, protection, monitoring and controlled utilization.

3.6 Silviculture

Definition — Silviculture: The branch of forestry concerned with the establishment, growth, composition, health and quality of forest stands through planned management.

Simplified silvicultural management cycle



3.7 Ancient fertilizers and pesticides

Traditional inputs may include composted organic residues, manures and plant-based preparations. Their value depends on composition, preparation and dose. “Natural” does not automatically mean harmless; evaluate efficacy and safety scientifically.

Exam focus: Unit III high-yield: compare polyhouse, hydroponics and terrace farming; explain drip irrigation; define organic farming, horticulture, forestry and silviculture; give management cycles and applications.